E-Commerce Sales Performance Analysis

Project Overview (Resume Version):

Analyzed an e-commerce sales dataset to evaluate business performance, customer purchasing behavior, and product profitability. Used Excel, SQL, and Power BI for data cleaning, analysis, and visualization to identify key insights such as top-performing products, profitable customers, regional sales trends, and the impact of discounts. The project supports data-driven decision making to improve sales strategy and business performance.

project Summary

- **Rows: 5,000**
- **Columns: 18**

Key Features:

- **Order Details: Order ID, Order Date, Order Status**
- **Customer Information: Customer Name, Region**
- **Product Details: Product Name, Category, Sub-Category**
- **Sales Metrics: Sales, Profit, Quantity, Discount**
- **Payment Information: Payment Mode (Credit Card, Debit Card, PayPal)**
- **Shipping Details: Ship Mode, Shipping Status**

Explore data analysis using SQL

- **Find total sales by payment mode.**

```
SELECT Payment_Method,
    SUM(Sales) AS Total_Sales
FROM tutorial_dp.coustmer
GROUP BY Payment_Method;
```

Payment_Method	Total_Sales
Credit Card	68871
Debit Card	16232
PayPal	13143

- **Find average discount given in each category.**

```
SELECT Category,
    AVG(Discount) AS Avg_Discount
FROM tutorial_dp.coustmer
GROUP BY Category;
```

Category	Avg_Discount
Technology	0.016666666666666673
Office Supplies	0.004166666666666667
Furniture	0.036666666666666646

- **Find products that generated negative profit (loss).**

```
SELECT Product_Name,
    SUM(Profit) AS Total_Profit
FROM tutorial_dp.coustmer
GROUP BY Product_Name
HAVING SUM(Profit) < 2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product_Name	Total_Profit
Report Cover	1
Manila Folder	1
Display Folder	1
Clear Folder	1
Report File	1

- **Find customer who generated highest sales.**

```
SELECT Customer_Name,
    SUM(Sales) AS Total_Sales
FROM tutorial_dp.coustmer
GROUP BY Customer_Name
ORDER BY Total_Sales DESC
LIMIT 1;
```

Result Grid | Filter Rows:

Customer_Name	Total_Sales
Stephen Strange	2978

- **Find total quantity sold for each product.**

```
SELECT Product_Name,
    SUM(Quantity) AS Total_Quantity
FROM tutorial_dp.coustmer
GROUP BY Product_Name
ORDER BY Total_Quantity DESC;
```

Result Grid | Filter Rows:

Product_Name	Total_Quantity
Pocket Folder	240
Notebook	65
Lined Paper	50
Scratch Paper	50
Notes	50

- **Find top 5 customers by total profit generated.**

```
SELECT Customer_Name,
    SUM(Profit) AS Total_Profit
FROM tutorial_dp.coustmer
GROUP BY Customer_Name
ORDER BY Total_Profit DESC
LIMIT 5;
```

Result Grid | Filter Rows:

Customer_Name	Total_Profit
Stephen Strange	590
James Rhodes	300
Galactus World	300
Scott Lang	295
Deadpool Wade	280

- **Find region with highest average sales per order**

```
SELECT Region,
    AVG(Sales) AS Avg_Sales_Per_Order
FROM tutorial_dp.coustmer
GROUP BY Region
ORDER BY Avg_Sales_Per_Order DESC
LIMIT 1;
```

Result Grid | Filter Rows:

Region	Avg_Sales_Per_Order
West	173.3090

- **Find most ordered product in each category top 6.**

```
SELECT Category,
    Product_Name,
    SUM(Quantity) AS Total_Quantity
FROM tutorial_dp.coustmer
GROUP BY Category, Product_Name
```

```
ORDER BY Category, Total_Quantity DESC
limit 6;
```

Category	Product_Name	Total_Quantity
Furniture	Desk Lamp	20
Furniture	Folding Stool	13
Furniture	Key Hooks	12
Furniture	Incense Holder	12
Furniture	Desk Art	12
Furniture	Wall Art	11

- **Find number of orders in each order status.**

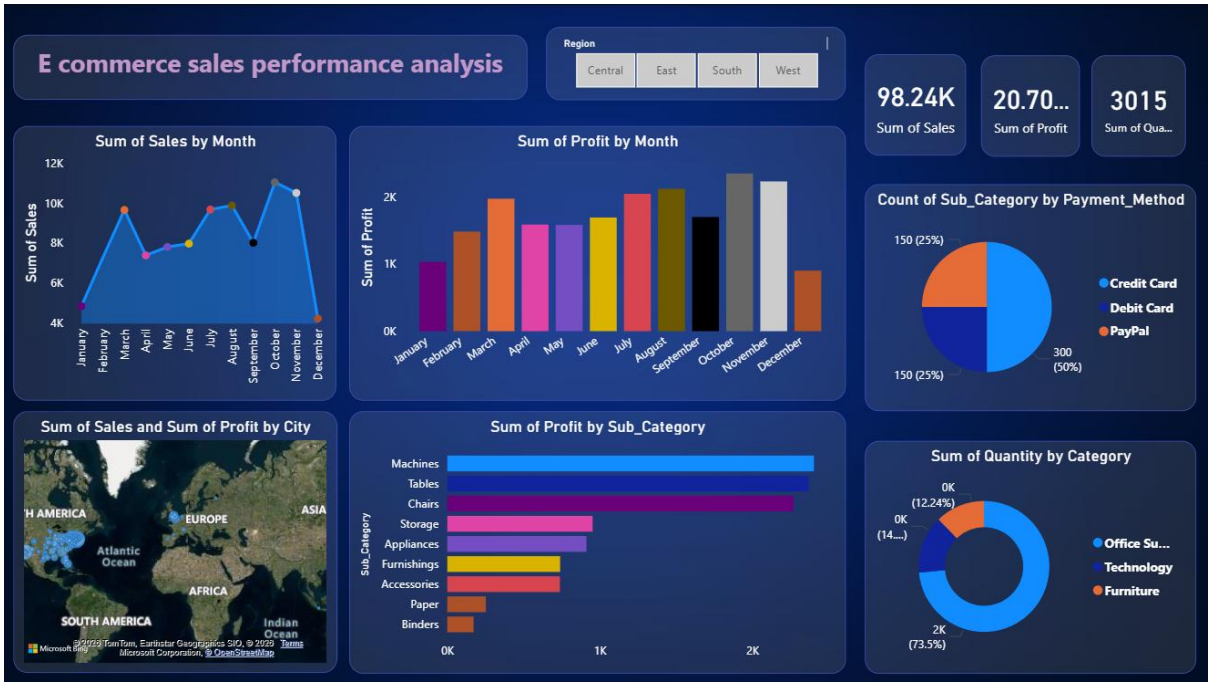
```
SELECT shipping_Status,
    COUNT(Order_ID) AS Total_Orders
FROM tutorial_dp.coustmer
GROUP BY shipping_Status;
```

shipping_Status	Total_Orders
Delivered	388
Shipped	152
Processing	60

- **Find monthly profit trend.**

```
SELECT MONTH(Order_Date) AS Month,
    SUM(Profit) AS Total_Profit
FROM tutorial_dp.coustmer
GROUP BY MONTH(Order_Date)
ORDER BY Month;
```

Month	Total_Profit
NULL	20732



➢ Business Recommendations

- **Improve High-Performing Products** – Focus marketing efforts on top-selling and high-profit products to increase revenue.
- **Reduce Loss-Making Products** – Identify products generating negative profit and adjust pricing or reduce discounts.
- **Optimize Discount Strategy** – Review discount policies to ensure sales growth without reducing overall profitability.
- **Strengthen Customer Retention** – Introduce loyalty programs and offers for repeat customers to increase long-term revenue.
- **Regional Sales Strategy** – Increase marketing campaigns in regions with lower sales and expand successful strategies from high-performing regions.

- **Improve Shipping Efficiency** – Reduce pending or delayed shipments to enhance customer satisfaction.
- **Promote Preferred Payment Methods** – Encourage commonly used payment methods to simplify transactions and improve customer experience.